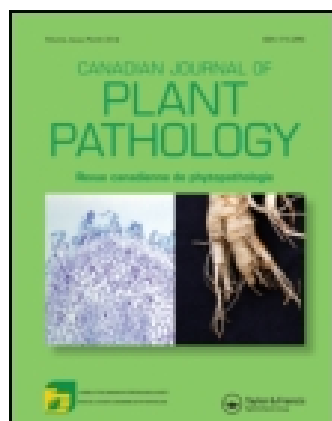




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Effect of soil temperature and inoculum density on pre-emergence damping-off of canola caused by *Rhizoctonia solani*

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Effect of soil temperature and inoculum density on pre-emergence damping-off of canola caused by *Rhizoctonia solani*

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The influence of temperature and of inoculum density of two isolates of *Rhizoctonia solani* on pre-emergence damping-off of canola cv. Westar was studied in the growth chamber. One isolate each of anastomosis group (AG) 2-1 and AG 4 was chosen for the study. Six inoculum densities, 0, 1400, 2800, 5700, 11 400, and 22 800 viable propagules/L soil-free growth medium, of each isolate were evaluated in the following six night-day air temperature regimes: 7-8, 7-12, 7-18, 12-18, 19-25, and 26-35°C. The corresponding soil temperatures were 4-8, 4-16, 5-22, 12-22, 17-26, and 25-36°C. Significant effects of temperature, isolate, inoculum density, and all possible interactions among these factors were demonstrated. At all inoculum densities tested in the 7-8 and 7-12°C regimes, and at the lower inoculum densities in the 7-18°C regime, the isolate of AG 2-1 caused more pre-emergence damping-off than the isolate of AG 4. However, the isolate of AG 4 caused more pre-emergence damping-off than the isolate of AG 2-1 at all inoculum densities in the 26-35°C regime and at the lower inoculum densities in the 12-18 and 19-25°C regimes. The two isolates caused similar levels of pre-emergence damping-off at the higher inoculum densities in the 7-18, 12-18, and 19-25°C regimes; increases in inoculum density of both isolates caused increases in pre-emergence damping-off below the level of the plateau.

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On a étudié en chambre de croissance l'effet de la température et de la densité d'inoculum de deux isolats de *Rhizoctonia solani* sur la fonte en pré-levée des semis du colza canola cv. Westar. Un isolat de chacun des groupes d'anastomose (AG) 2-1 et 4 a été choisi pour cette étude. On a comparé six densités d'inoculum, soit 0, 1400, 2800, 5700, 11 400, et 22 800 propagules par litre de milieu de culture sans sol, sous les six régimes de température nuit-jour suivants: 7-8, 7-12, 7-18, 12-18, 19-25, et 26-35°C. Les températures du sol correspondantes étaient 4-8, 4-16, 5-22, 12-22, 17-26, et 25-36°C. L'étude a fait ressortir des effets significatifs pour la température, l'isolat, la densité d'inoculum et toutes les interactions possibles de ces facteurs. Pour toutes les densités d'inoculum, aux régimes thermiques 7-8 et 7-12°C, et pour les basses densités au régime 7-18°C, l'isolat de AG 2-1 a provoqué plus de fonte des semis que l'isolat de AG 4. Toutefois cet isolat a causé plus de fonte que celui de AG 2-1 au régime 26-35°C pour toutes les densités d'inoculum, et aux régimes 12-18 et 19-25°C aux basses densités. Les deux isolats ont suscité des intensités comparables de maladie aux fortes densités dans les régimes 7-18, 12-18, et 19-25°C. L'accroissement de la densité d'inoculum des deux isolats a donné lieu à une augmentation de la fonte de pré-levée en-dessous du niveau du plateau.

Rhizoctonia solani Kühn is a major root pathogen of canola (*Brassica napus* L. and *B. campestris* L.) in western Canada (7, 10, 11, 20). Anastomosis group (AG) 2-1 and AG 4 are most frequently isolated from diseased plants in this region. When plants are artificially inoculated, these are the only AGs which cause symptoms similar to the symptoms of post-emergence damping-off and root rot in infested fields (7, 11, 20).

Kaminski and Verma (11) demonstrated that maximum hyphal growth rates of isolates of AG 2-1 and AG 4 occurred at different temperatures and that the temperature ranges allowing growth differed among AGs. However, the virulence of isolates was not investigated over such temperature ranges.

In a constant environment, an increase in inoculum density (ID) of *R. solani* is expected to cause an increase in disease incidence (DI) or severity until a plateau is reached. Although well documented for radish (3) and beans (18, 21), ID-

DI relationships have not been studied for *R. solani* on canola.

The main objective of the present study was to investigate the influence of temperature on the virulence of two isolates of *R. solani* in causing pre-emergence damping-off of canola. Virulence was studied over a range of inoculum densities; hence, the interaction of temperature and inoculum density enabled the detection of differences in ID-DI relationships at different temperatures.

Materials and methods

Inoculum of *R. solani* AG 2-1 (isolate no. 4) and AG 4 (isolate no. 35) (11) was prepared using procedures described previously (1,7,11). Each isolate was grown on moist autoclaved rye kernels (100 mL rye kernels + 150 mL water, soaked for 6-12 h and autoclaved for 1 h) for 2 weeks at 25°C, air-dried in the dark for 48 h, ground and sieved for particles of 0.6-1.2 mm diameter. Viability of the inoculum was tested by first weighing and then

planting 100 randomly selected particles of each isolate on Difco potato dextrose agar amended with 0.001% streptomycin sulfate. This information was used to calculate the weight of ground inoculum required per litre of soil-free growth medium (SFM) (16) to provide the following inoculum densities: 0 (= control), 1400, 2800, 5700, 11 400, and 22 800 viable propagules (VP)/L SFM. The constituents of the SFM were as follows: 1 part peat moss, 1 part fine vermiculite, and 1/2 part medium grade sand. For every 36 L of this mix, 135 g ground limestone, 70 g of 20% superphosphate, 120 g control release fertilizer 18-6-12, 1 g iron chelate, and 2 g fritted trace elements were added (16).

The six inoculum densities described above were evaluated in a growth chamber over six air temperature regimes chosen to approximate a range of spring growing conditions. The six air temperature regimes were 7-8, 7-12, 7-18, 12-18, 19-25, and 26-35°C (nightly minimum-daily maximum). Corresponding soil temperatures, measured at a depth of 2-3 cm with a Weksler 7-day soil thermograph, were 4-8, 4-16, 5-22, 12-22, 17-26, and 25-36°C, respectively. Discrepancies between corresponding soil and air temperatures are attributable to differences in equipment used to measure soil and air temperatures, and to the circulation of air in the growth chamber, which is designed in such a way that cool air flows beneath the pots thereby causing lower soil temperatures. The growth chamber was programmed for a 16 h photoperiod with a constant light intensity of $371 \mu\text{E m}^{-2}\text{s}^{-1}$. Changes in air temperature from the nightly minimum to the daily maximum were programmed to occur at a rate of 1°C/h except for the 7-18 and 26-35°C regimes, where the rate of change was 2°C/h.

The experimental design was a 6 x 6 x 2 factorial in a randomized complete block with each treatment replicated in 10 pots. For each temperature regime, 25 seeds of *B. napus* cv. Westar were planted on the lightly packed surface of SFM in 10-cm-diameter pots. The seeds were covered with 100 mL of SFM infested with one of the *R. solani* isolates at one of the six inoculum densities described previously. Infestation of the SFM was accomplished by hand-mixing the appropriate amounts of inoculum and SFM.

Seedling emergence was recorded 10 days after planting except for the 7-8°C regime, which was recorded at day 12. Mean values of percent pre-emergence damping-off were plotted against inoculum density. Since the standard errors of the means were all very small, the pooled standard error was calculated for the experiment. The data

were then transformed with $\text{ARCSINE } \sqrt{\%/100}$ to improve the equality of the variances and the effects of the experimental factors on pre-emergence damping-off were tested using analysis of variance of the transformed data.

To verify the results obtained, the experiment was repeated using only three air temperature regimes, 7-8, 12-18, and 26-35°C.

Results and discussion

Significant ($P = 0.01$) effects of temperature, isolate, and inoculum density were demonstrated on percent pre-emergence damping-off of canola cv. Westar in both experiments (Table 1). All possible interactions among these factors were also significant. In the first experiment, at all inoculum densities tested in the 7-8 and 7-12°C regimes and at inoculum densities of 5700 VP/L SFM or less in the 7-18°C regime, the isolate of AG 2-1 was more virulent than the isolate of AG 4, causing higher levels of pre-emergence damping-off (Fig. 1). However, the isolate of AG 4 was more virulent than the isolate of AG 2-1 at all inoculum densities in the 26-35°C regime and at inoculum densities of 5700 VP/L SFM or less in the 12-18 and 19-25°C regimes. Similar levels of pre-emergence damping-off were caused by both isolates at the two highest inoculum densities in the 12-18 and 19-25°C regimes (Fig. 1). The second experiment confirmed these findings (Fig. 2).

Table 1. Analysis of variance of the influence of temperature (TEMP), isolate of *Rhizoctonia solani* (ISOL), and inoculum density (ID) of *R. solani* on percent pre-emergence damping-off of *Brassica napus* cv. Westar in two growth chamber experiments

Source of variation	DF	Mean square
Test 1:		
Replication	9	27.54
TEMP	5	27395.36**
ISOL	1	3113.72**
ID	5	82126.29**
TEMP x ISOL	5	30373.13**
TEMP x ID	25	1477.65**
ISOL x ID	5	234.58**
TEMP x ID x ISOL	25	1765.91**
Error	639	25.87
Test 2:		
Replication	9	26.81
TEMP	2	23051.04**
ISOL	1	1037.86**
ID	5	31241.18**
TEMP x ISOL	2	33372.77**
TEMP x ID	10	1801.43**
ISOL x ID	5	161.27**
TEMP x ID x ISOL	10	1720.42**
Error	315	18.26

**Significant at $P = 0.01$.

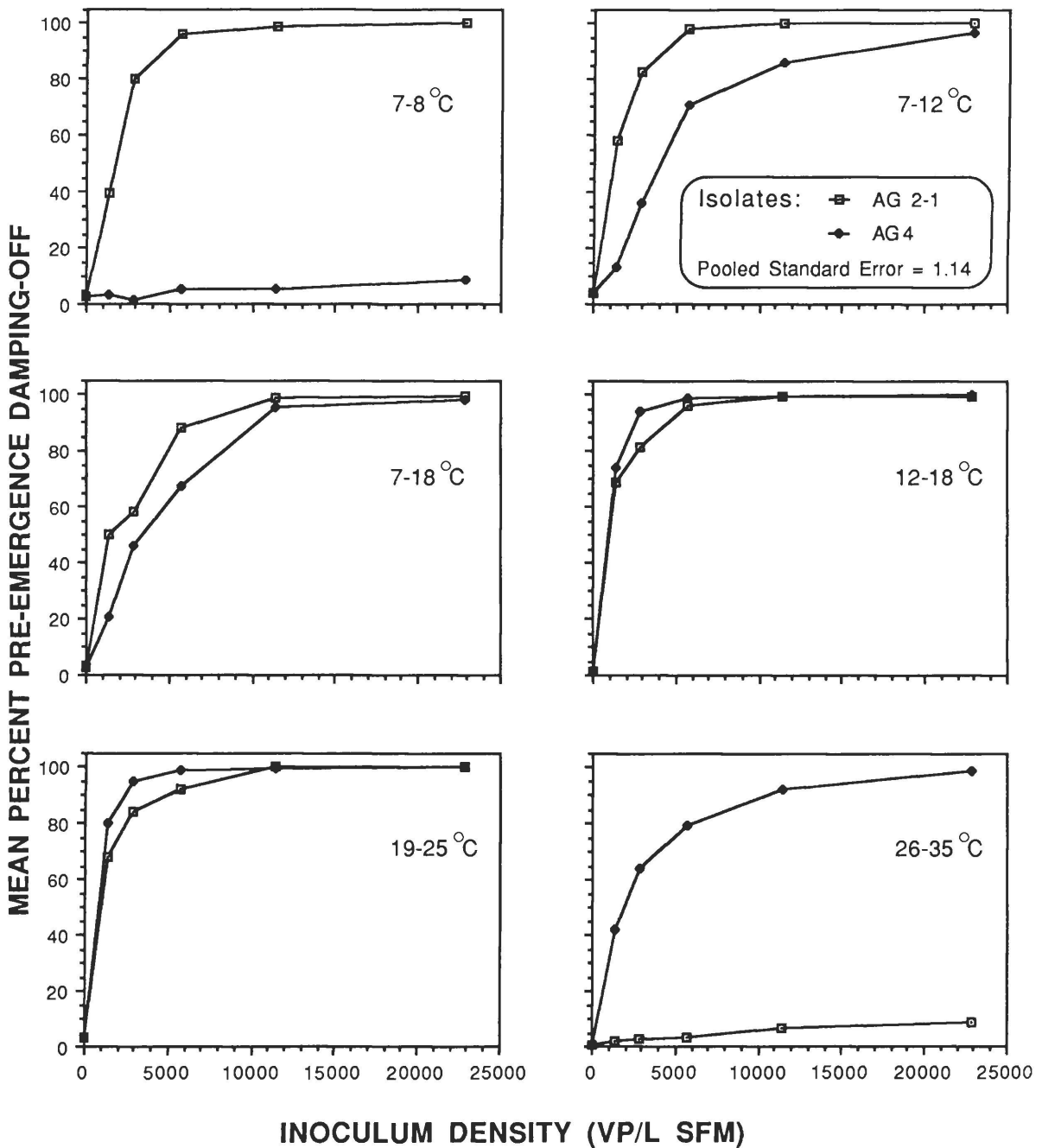


Figure 1. Effect of air temperature, isolate of *Rhizoctonia solani*, and inoculum density of *R. solani* on percent pre-emergence damping-off of *Brassica napus* cv. Westar in Experiment I.

Both experiments demonstrated that increases in inoculum density were associated with increases in disease incidence. This has also been shown for *R. solani* on radish (4, 13), dry beans (18), and

soybeans (21). The two isolates of *R. solani* used in this study exhibited a considerable difference in their response to temperature. The isolate of AG 2-1 was more virulent than the isolate of AG 4 at

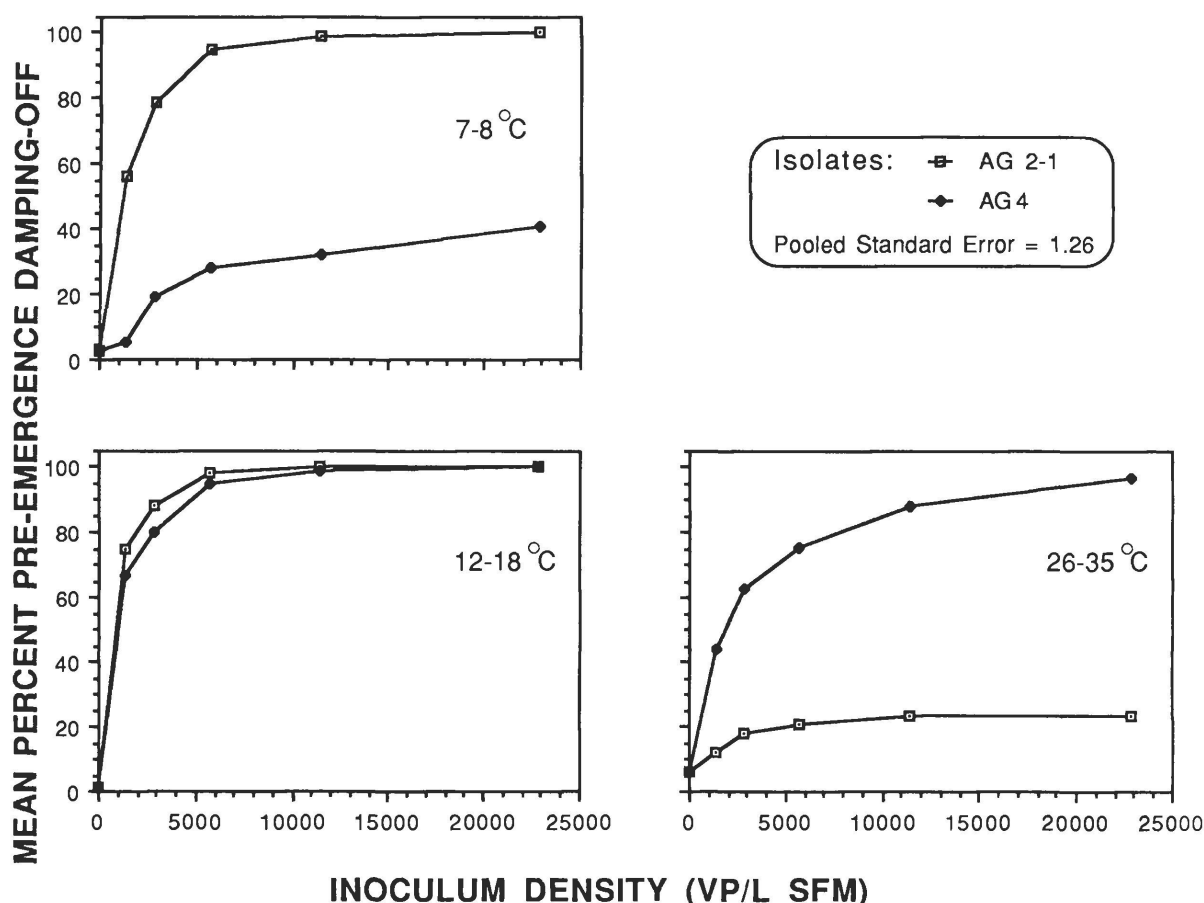


Figure 2. Effect of air temperature, isolate of *Rhizoctonia solani*, and inoculum density of *R. solani* on percent pre-emergence damping-off of *Brassica napus* cv. Westar in Experiment 2.

relatively low temperatures, and the isolate of AG 4 was more virulent than the isolate of AG 2-1 at relatively high temperatures.

The effect of temperature on the virulence of the two isolates in the present study may be partly related to their mycelial growth rates. The optimum temperatures for in vitro mycelial growth of canola isolates of AG 2-1 and AG 4, respectively, were 24 and 26°C (11). Isolates of AG 2-1 grew at 2°C but not at 36°C, whereas the reverse was true for isolates of AG 4. However, it is not immediately obvious how a 2°C differential in the optimum temperature for growth of these two isolates can account for the great difference in pre-emergence damping-off they caused at the temperature extremes in this study. Shehata et al. (14) and van Bruggen et al. (18) suggested that the effect of soil temperature on infection by *R. solani* may be related to changes in host vigour at different

temperatures. In the present study, about 80% seedling emergence was observed in the uninoculated pots at 26-35°C within four days of planting, whereas 10-12 days were required to achieve the same level of emergence at 7-8°C. Beach (2) and Leach (12) concluded that the severity of pre-emergence infection at different temperatures was largely determined by the relative growth rates of both host and pathogen. In contrast, Doornik (6) suggested that mycelial growth rate was not the only factor influencing infection, and that the optimum temperature for pathogenicity is an intrinsic characteristic of the pathogen. Further research is required to resolve this conflict.

A similar relationship of virulence with temperature was also observed in field studies with the same two isolates of *R. solani* used in the present study. An experiment was conducted in 1986 using two seeding dates four weeks apart in an

attempt to obtain different soil temperature regimes during the emergence and seedling stages of the crop (19). No significant difference in virulence between the two isolates was observed, probably because both seeding dates were when the soil temperature was in the range of 15-25°C; within this range the two isolates demonstrated similar virulence in the growth chamber. However, in another field study, planting canola before the soil had warmed to 15°C was more favourable to the isolate of AG 2-1 than to the isolate of AG 4 in causing pre- and post-emergence damping-off (17). Disease due to the isolate of AG 4 increased as the soil warmed to 20°C, which is consistent with studies on radish (4), pea (14), and parsley (8). Pre- and post-emergence damping-off of parsley caused by *R. solani* AG 4 was more severe at 23-30°C than at 15°C both in sand beds and in growth chamber tests. Pre-emergence damping-off of radish was more severe at 26°C than at 15, 20, 22, or 30°C; 6000-10 000 propagules/kg soil were required to cause 20-80% DI at 15°C, compared with only 500-1500 propagules/kg soil at 26°C (4). The virulence at relatively low soil temperatures of the isolate of AG 2-1 on canola in the present study was similar to the reported virulence of other isolates of AG 2-1 on radish (9). Isolates of AG 2-1 and AG 4 were the most frequent isolates from radish. However, AG 2-1 was obtained most frequently from plants grown in winter or at low temperatures during autumn, while AG 4 was obtained most frequently from plants grown at higher temperatures. Pathogenicity tests demonstrated increased virulence of isolates of AG 2-1 and AG 4 at low and high temperatures, respectively (9).

The increased virulence of isolates of AG 2-1 over AG 4 at relatively low soil temperatures and of isolates of AG 4 over AG 2-1 at higher soil temperatures is reflected in the frequency with which these AGs were isolated from diseased seedlings and mature plants in Saskatchewan (20). Isolates of *R. solani* obtained from seedlings were almost exclusively assigned to AG 2-1. However, AG 4 was isolated almost five times more frequently from mature plants than from seedlings. Soil temperatures tend to be low in the spring and to increase as the growing season progresses. The high virulence of the isolate of AG 2-1 at relatively low temperatures, in addition to the prevalence of AG 2-1 in Peace River soils (19), is probably one reason why root rot of canola is a serious problem in the Peace River region of Alberta (5, 15).

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